ClassVibe - Hybrid Multimodal Speech Emotion Recognition using Hubert and Handcrafted Acoustic Features on RAVDESS

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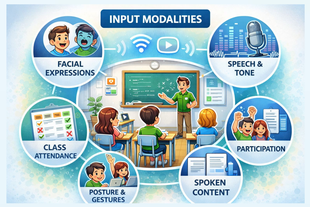
Introduction

Effective teaching is not determined solely by the delivery of content, but also by the emotional dynamics that emerge during a lecture. In a classroom, both students and teachers continuously express affective cues through facial expressions, voice, tone, engagement, and behavioral reactions. These emotional signals can indicate whether a lecture is stimulating curiosity, maintaining attention, encouraging participation, or becoming emotionally flat and ineffective. Lectures associated with positive and productive emotions, such as interest, curiosity, attentiveness, and satisfaction, are often more beneficial for student learning than lectures characterized by boredom, confusion, or emotional disengagement.

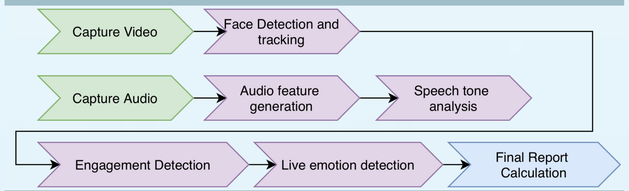
With recent advances in artificial intelligence, computer vision, and speech analysis, it is increasingly possible to study classroom interactions through multimodal data. Video can provide facial expressions, gestures, and visual engagement cues, while audio can capture tone, pitch, energy, and other vocal indicators of emotion. However, building a practical system for classroom emotion understanding remains a challenging task. In real classroom environments, multiple students may appear simultaneously, faces must be detected and separated correctly, speakers may overlap, background noise may interfere with audio analysis, and emotional cues may be subtle, mixed, or continuously changing over time. These challenges become even more complex when the goal is to perform analysis in real time during a live lecture rather than as an offline post-processing task.

This project aims to develop a **multimodal AI system for live classroom emotion detection** using audio and, when available, video streams. The system is designed to identify emotional patterns expressed by students and teachers, estimate the overall emotional state of the lecture, and assess whether the classroom environment is emotionally engaging or emotionless. By combining speech-based emotion recognition with visual emotion understanding, the proposed framework seeks to provide a more complete representation of classroom affect. Such a system could support educators by offering immediate feedback about classroom engagement, emotional response, and lecture effectiveness, thereby enabling responsive teaching strategies.

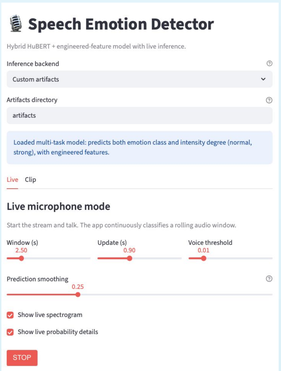
At its core, this research addresses the challenge of transforming complex, real-time classroom signals into meaningful measures of emotional engagement. The broader motivation is not only to detect emotions, but also to understand how emotional patterns within lectures relate to teaching quality, student attentiveness, and the overall learning experience.

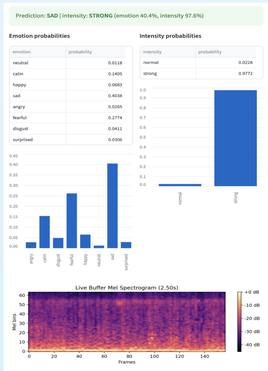


Application Overview

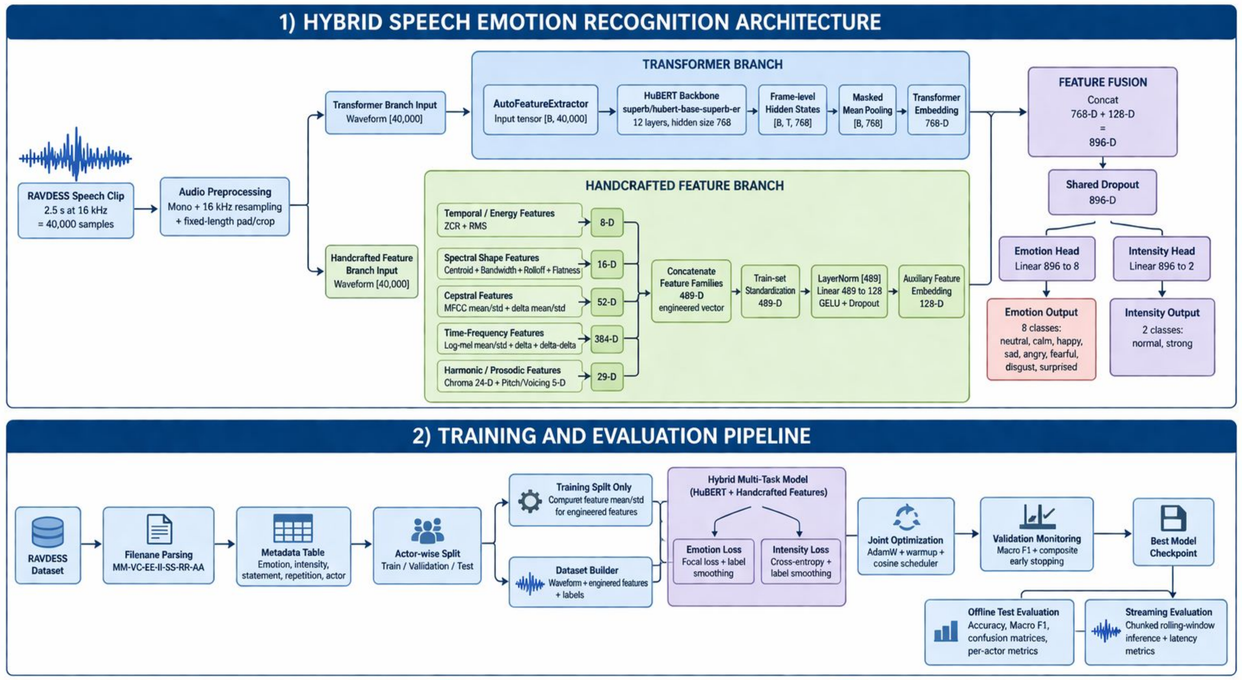


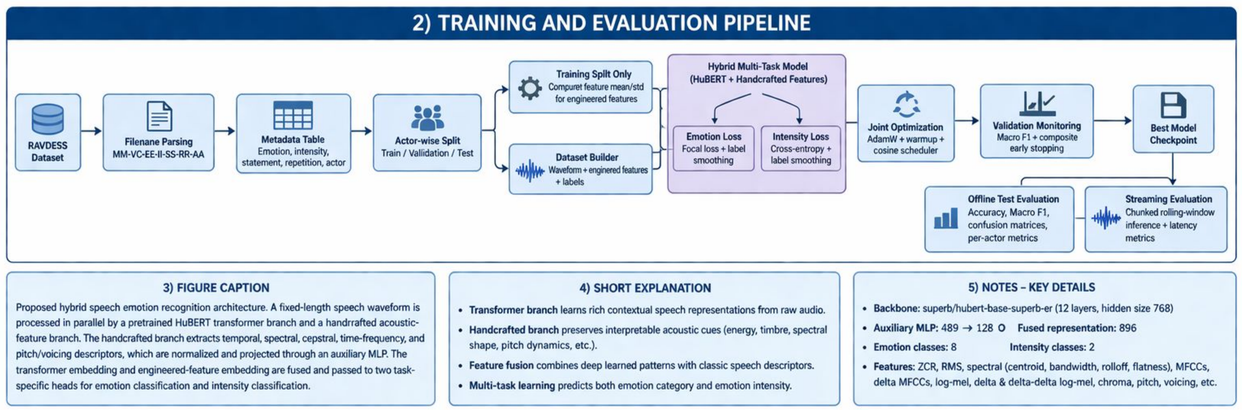
Application Interface





Methodology





3) FIGURE CAPTION

Proposed hybrid speech emotion recognition architecture. A fixed-length speech waveform is processed in parallel by a pretrained HuBERT transformer branch and a handrrafted acoustic-feature branch. The handcrafted branch extracts temporal, spectral, cepstral, time-frequency, and pitch/voicing descriptors, which are normalized and projected through an auxiliary MLP. The transformer embedding and engineered-feature embedding are fused and passed to two task-specific heads for emotion classification and intensity classification.

4) SHORT EXPLANATION

- **Transformer branch** learns rich contextual speech representations from raw audio.
- **Handcrafted branch** preserves interpretable acoustic cues (energy, timbre, spectral shape, pitch dynamics, etc.).
- **Feature fusion** combines deep learned patterns with classic speech descriptors.
- **Multi-task learning** predicts both emotion category and emotion intensity.

5) NOTES – KEY DETAILS

- **Backbone:** superb/hubert-base-superb-er (12 layers, hidden size 768)
- **Auxiliary MLP:** 489 → 128 O **Fused representation:** 896
- **Emotion classes:** 8 **Intensity classes:** 2
- **Features:** ZCR, RMS, spectral (centroid, bandwidth, rolloff, flatness), MFCCs, delta MFCCs, log-mel, delta & delta-delta log-mel, chroma, pitch, voicing, etc.

Diagram of the proposed methodology

Data Processing

- Use the RAVDESS dataset as the main source of labeled emotional speech data.
- Focus on speech samples with annotated emotion and intensity labels.
- Use the standardized dataset structure for reliable training and evaluation.

Feature Engineering

- Extracted a 489-dimensional handcrafted acoustic feature vector from each speech clip.
- Converted audio to mono, 16 kHz, and a fixed 2.5-second duration.
- Included ZCR and RMS statistics (8D).
- Included spectral centroid, bandwidth, rolloff, and flatness statistics (16D).
- Included MFCC and delta-MFCC mean/std features (52D).
- Included log-mel, delta log-mel, and delta-delta log-mel mean/std features (384D).
- Included chroma mean/std features (24D).
- Included pitch statistics and voiced ratio (5D).
- Standardized the full feature vector using train-split mean and standard deviation.
- Projected features through a 489 -> 128 auxiliary MLP.
- Fused the 128-dimensional handcrafted feature embedding with the 768-dimensional HuBERT embedding.
- Used the fused representation for hybrid multi-task emotion and intensity classification.

Model Architecture

- **Hybrid architecture** with **HuBERT + handcrafted acoustic features**
- **HuBERT backbone** generates a **768-D speech embedding**
- **489-D handcrafted features** are projected to **128-D**
- Both embeddings are **concatenated into an 896-D fused vector**
- Two output heads perform:
  - **8-class emotion classification**
  - **2-class intensity classification**

Evaluation & Real-Time Application

- Test model performance on **emotion classification** and **intensity classification**.
- Report **accuracy, macro F1, balanced accuracy, joint accuracy, and confusion matrices**.
- Evaluate generalization using an **actor-wise train/validation/test split**.
- Measure **streaming performance and latency** using chunked rolling-window inference.
- Deploy a **Streamlit-based interactive app** for live and clip-based emotion prediction.
- Demonstrate **real-time emotion and intensity detection** from speech input.

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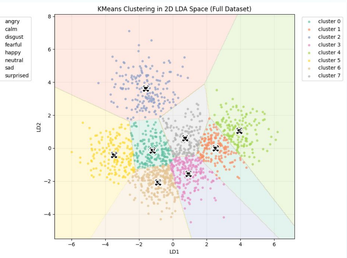
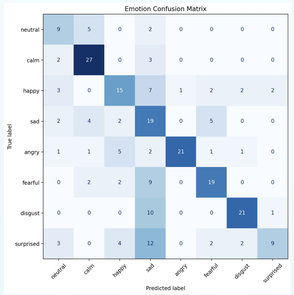
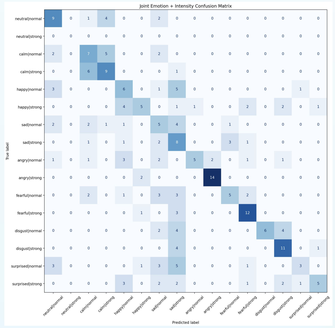
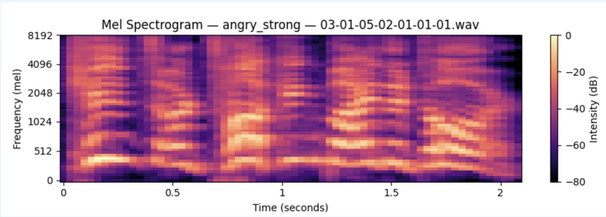
Results







Sample of Augmentation



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